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Department of CSE

B.Tech. CSE Trimester -5

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Cloud Infrastructure and Services

25CS2206E

LAB RECORD

2026

Experiment -1

AIM: IAM User Creation, Access Key Generation and AWS Billing Alert Configuration.

Description:

AWS Identity and Access Management (IAM) is used to create and manage users and their permissions securely. In this experiment, an IAM user is created with programmatic access only, and Access Key ID and Secret Access Key are generated and downloaded for accessing AWS services through the CLI.

AWS Free Tier usage alerts are enabled through Billing Preferences to receive email notifications when usage approaches or exceeds the specified limits.

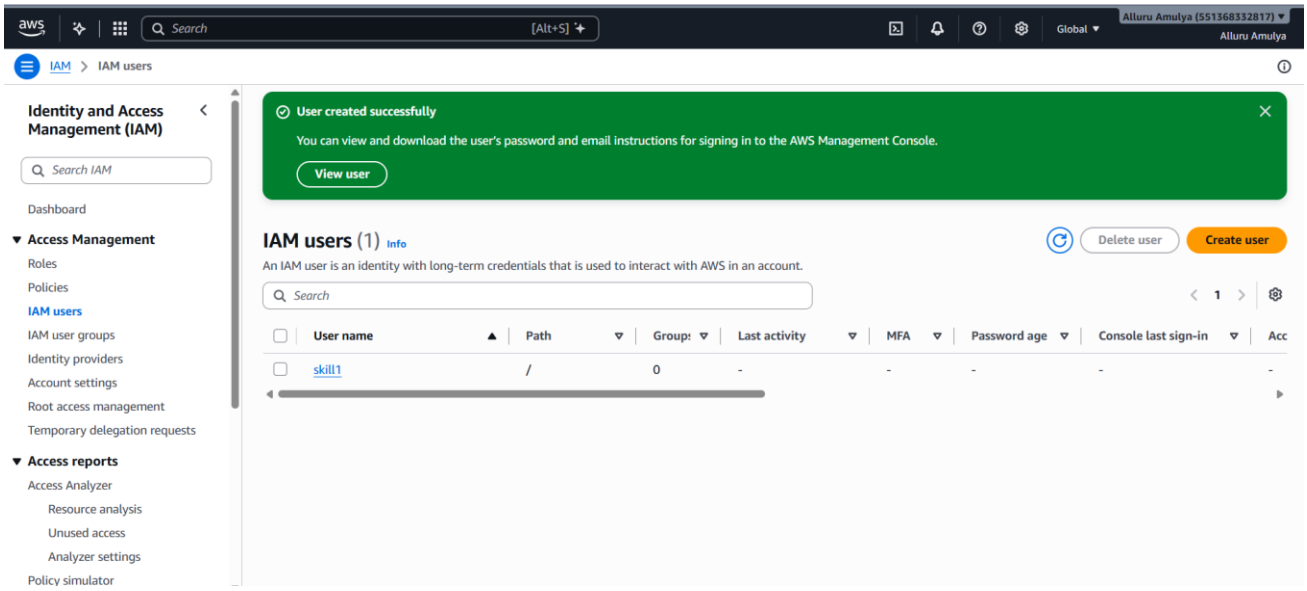
Additionally, an AWS monthly cost budget is created with email alerts to monitor spending and avoid unexpected charges. This experiment helps understand secure user management, programmatic access, and cost monitoring in AWS.

Steps:

Part A: Create an IAM User with Programmatic Access Only

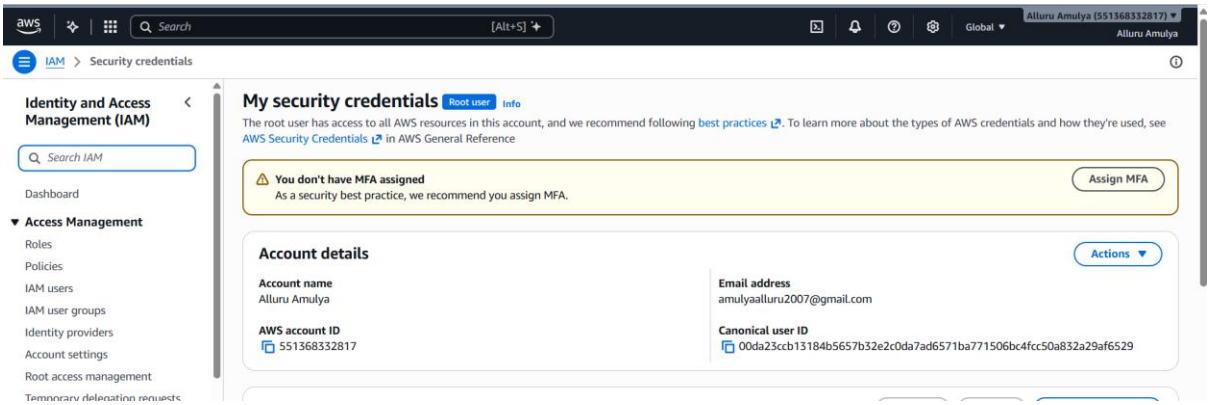
1. Log in to the AWS Management Console using your root account credentials.
2. In the search bar at the top of the AWS Console, search for IAM and select Identity and Access Management (IAM).
3. From the left navigation panel, select Users.
4. Click Create user.
5. Enter a suitable username (e.g., Student-IAM-User).
6. Do not enable AWS Management Console access, as the user requires programmatic access only.

7. Click Next.
8. Under Set permissions, select Attach policies directly.
9. Select the required permissions policy based on the tasks to be performed. For example, select Administrative access based on the tasks.
10. Click Next and review the configuration.
11. Click Create user to complete the process.
12. Verify that the newly created IAM user appears in the Users list.

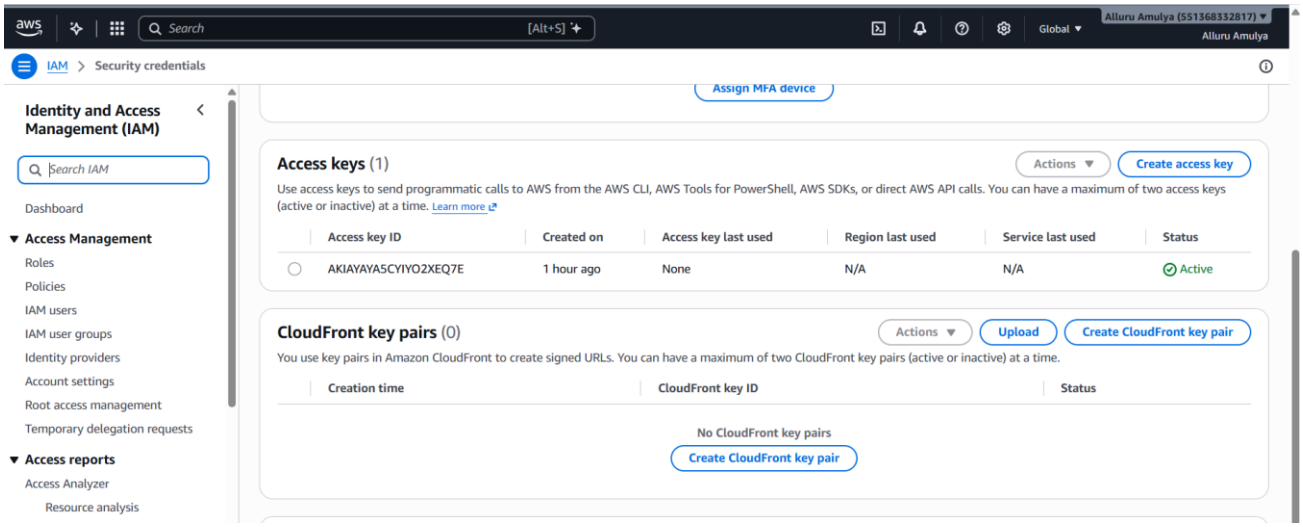


Part B: Generate and Download Access Keys

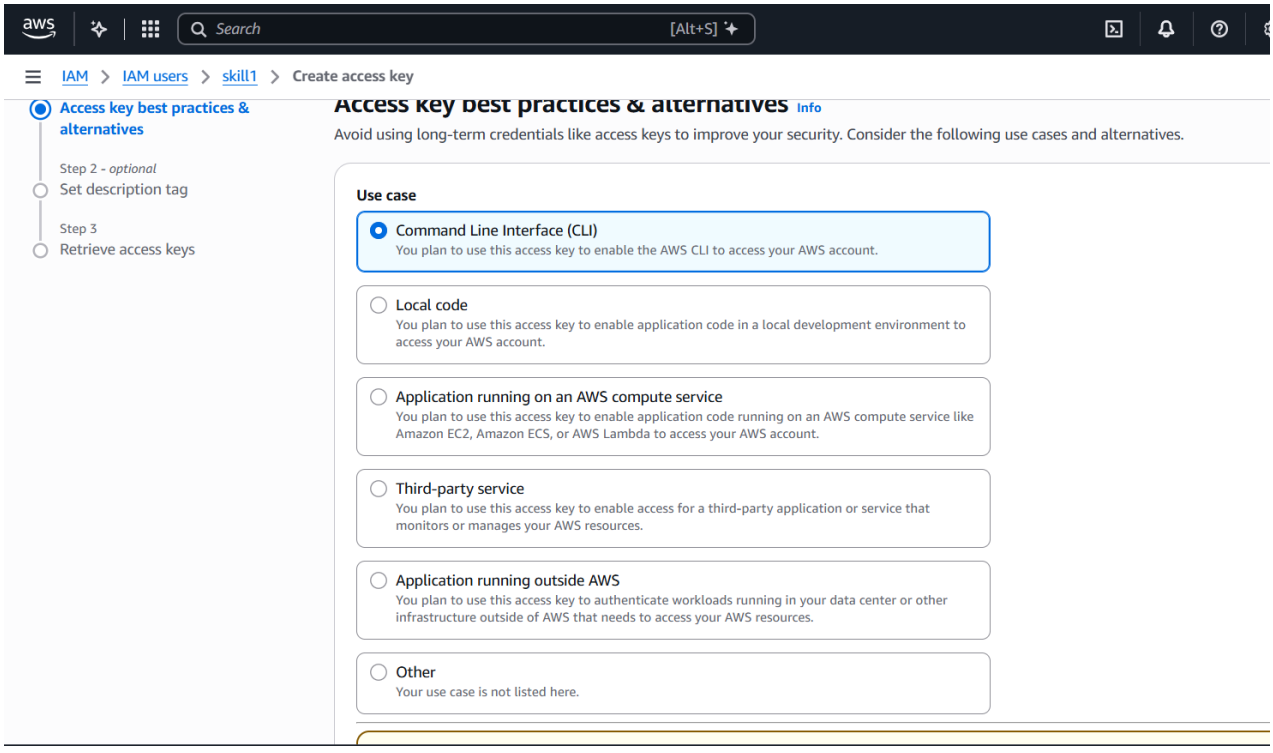
1. In the IAM dashboard, select the newly created IAM user.
2. Open the Security credentials tab.



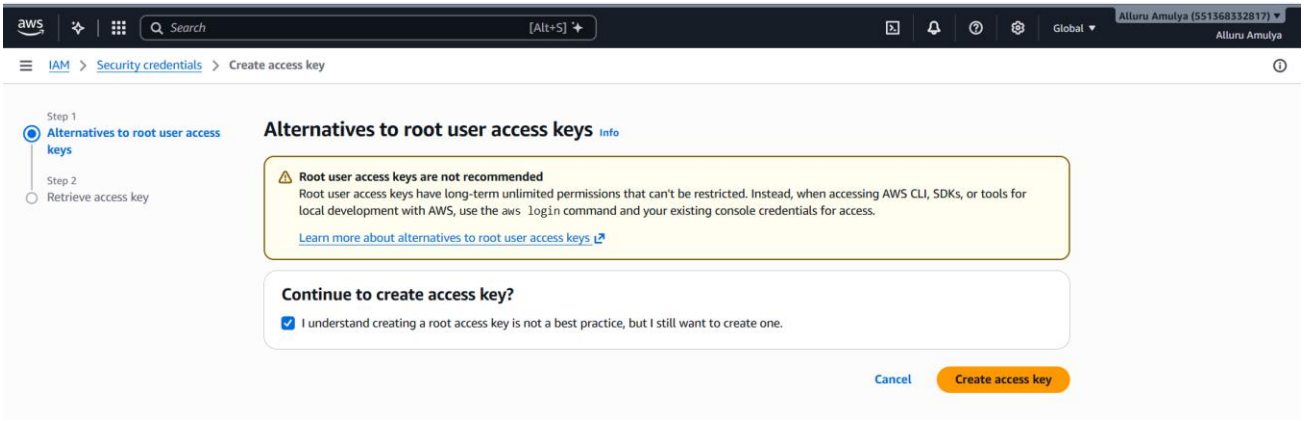
3. Scroll down to the Access keys section.
4. Click Create access key.



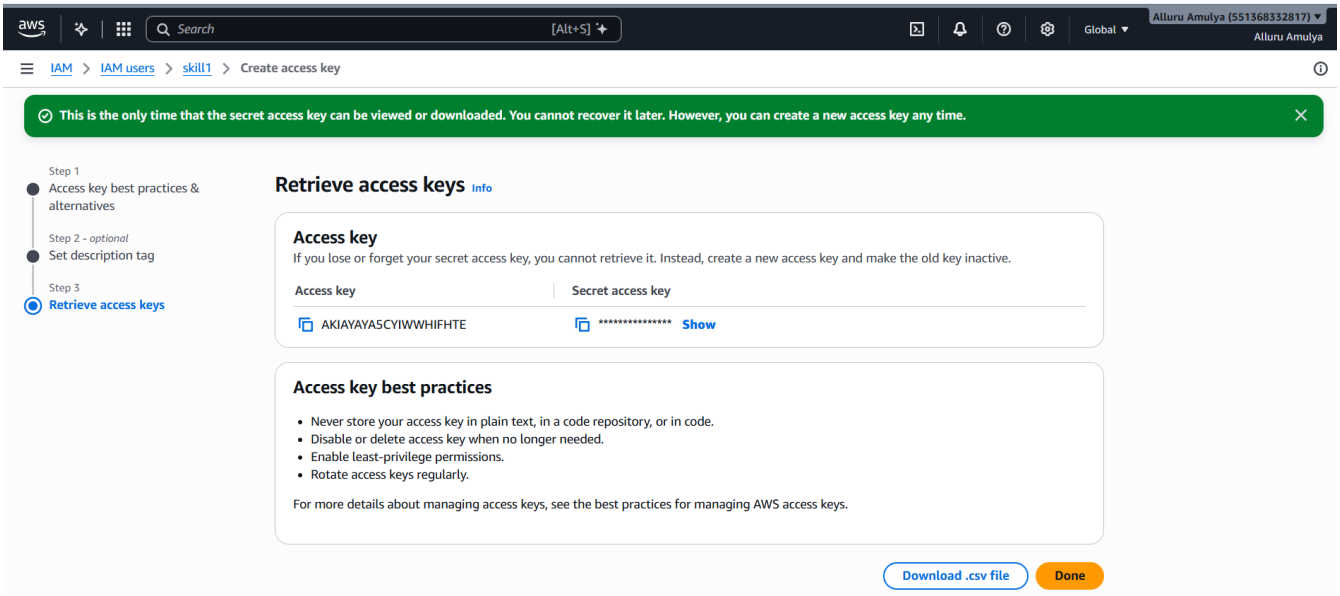
5. Select the appropriate use case, such as Command Line Interface (CLI), if the key is intended for AWS CLI access.
6. Acknowledge the recommendation and click Next.



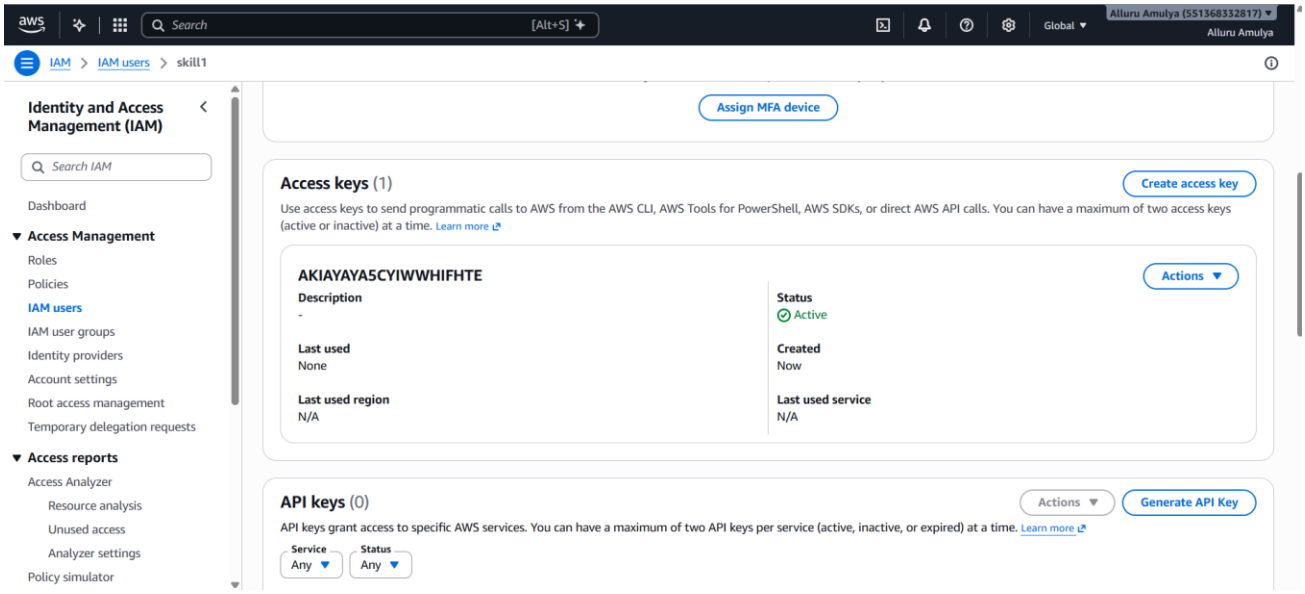
7. Optionally enter a description tag (e.g., Student-CLI-Access-Key).
8. Click Create access key.



9. The console displays the following credentials:
    - Access Key ID
    - Secret Access Key



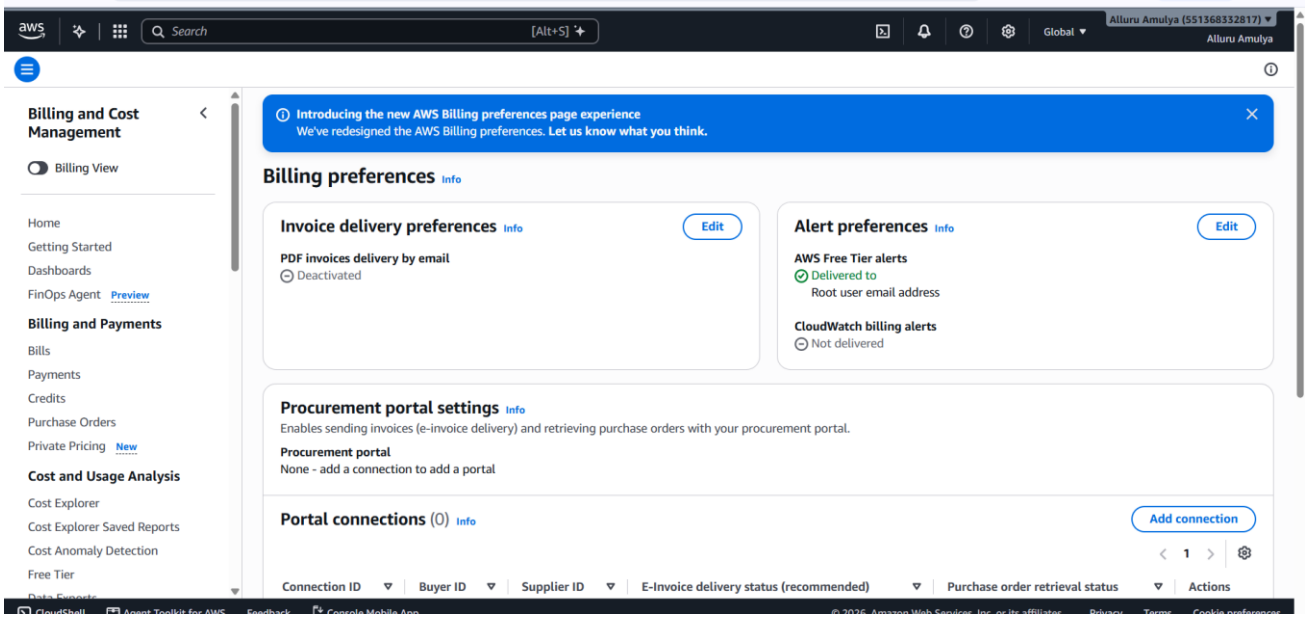
10. Click Download .csv file to download the credentials securely.
11. Verify that the downloaded CSV file contains both the Access Key ID and Secret Access Key.
12. Click Done.



Part C: Configure AWS Free Tier Usage Alerts

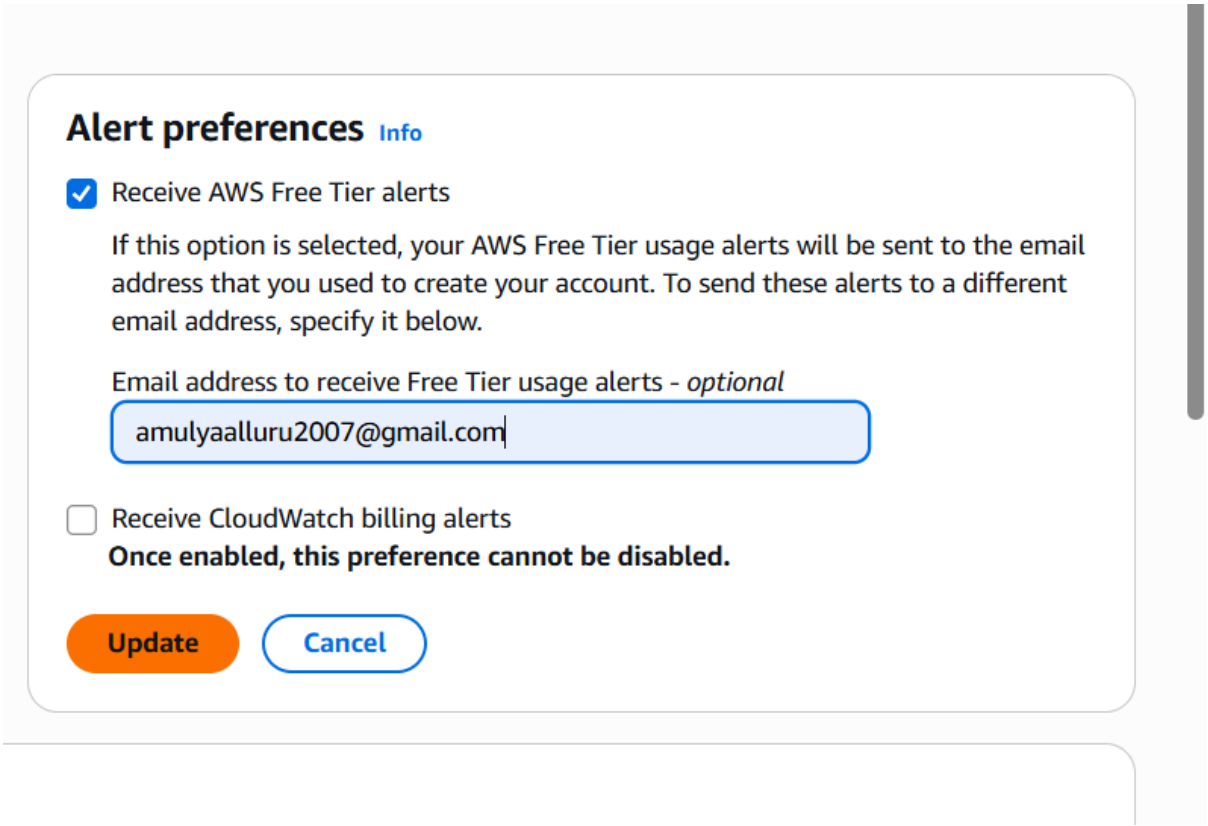
Configure alerts through Billing Preferences.

1. Log in to the AWS Management Console using the Root User account.
2. Search for Billing and Cost Management in the AWS search bar and open it.
3. In the left navigation panel, locate Preferences and select Billing preferences.

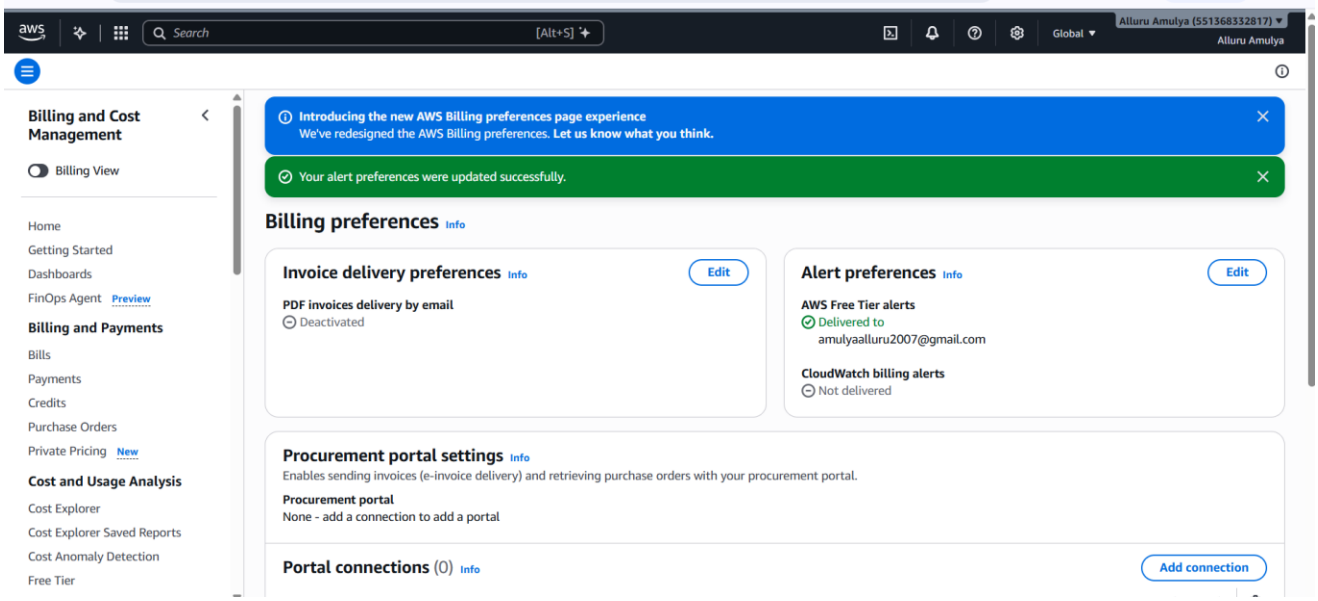


4. Locate the Alert preferences section.
5. Click Edit.

6. Select the checkbox Receive AWS Free Tier usage alerts.
7. Under Additional email address to receive alerts, enter the email address where notifications should be received (if required).

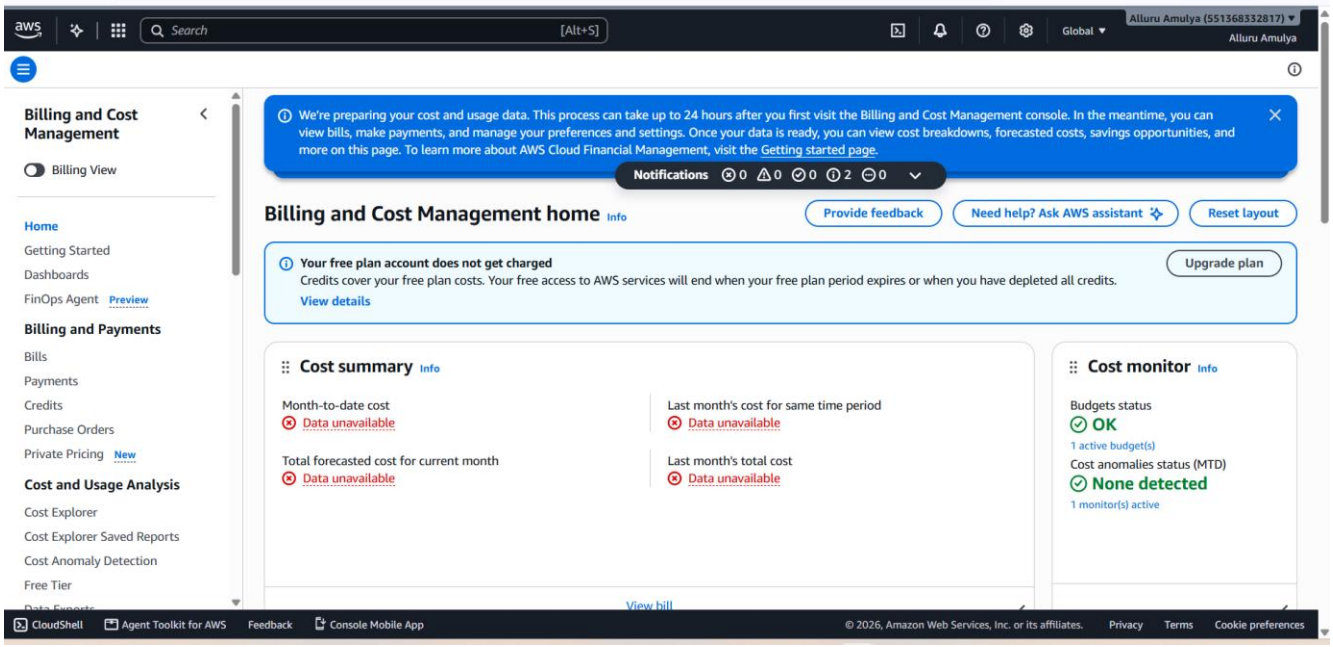


8. Click Update to save the settings.
9. Verify that the Free Tier usage alert option is enabled.

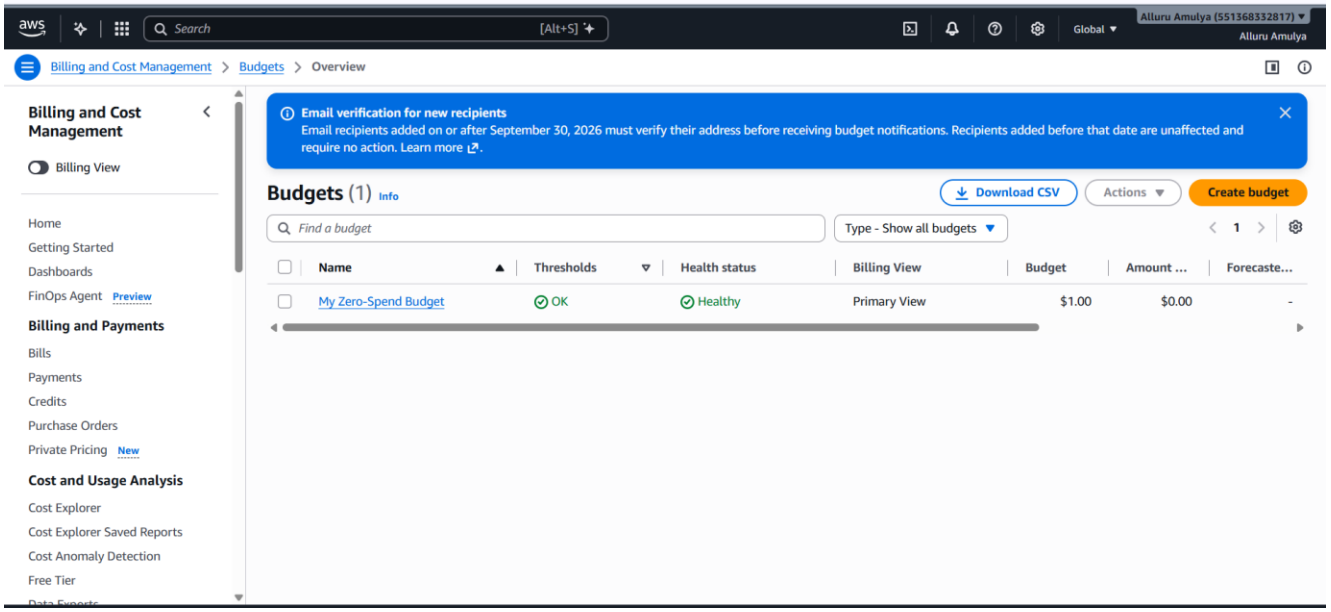


Part D: Configure AWS Billing Alerts Using the Custom Budget Option

1. Log in to the AWS Management Console using the root account.
2. Search for Billing and Cost Management and open it.

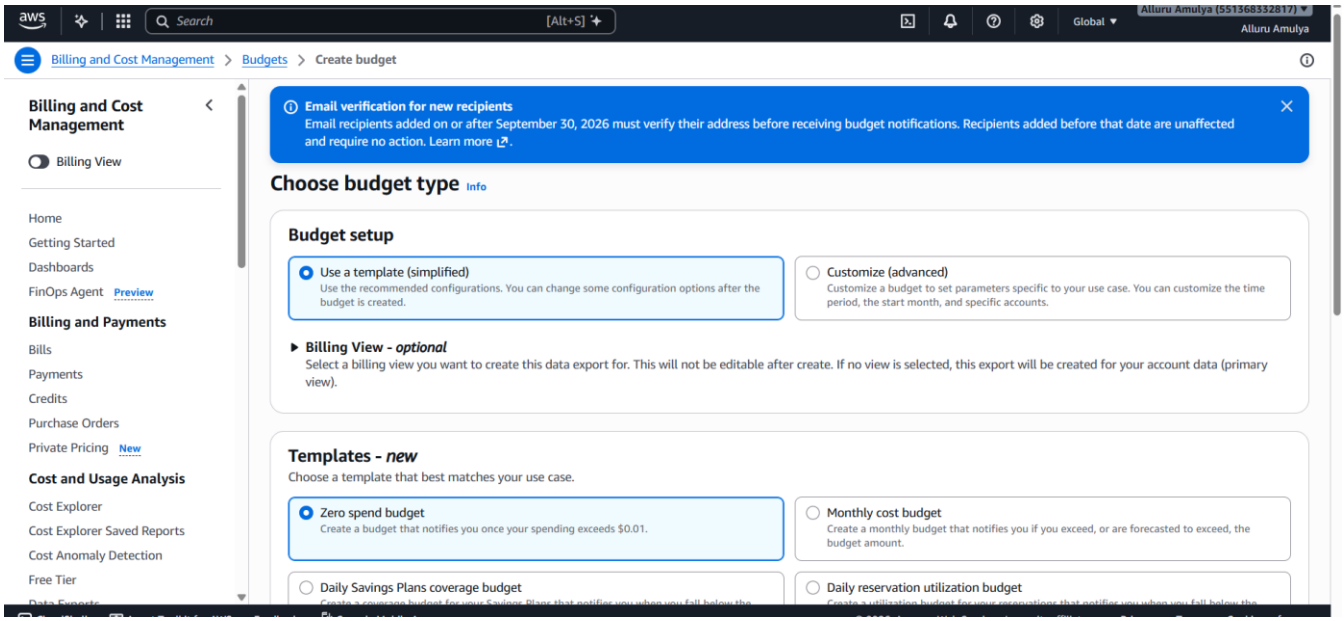


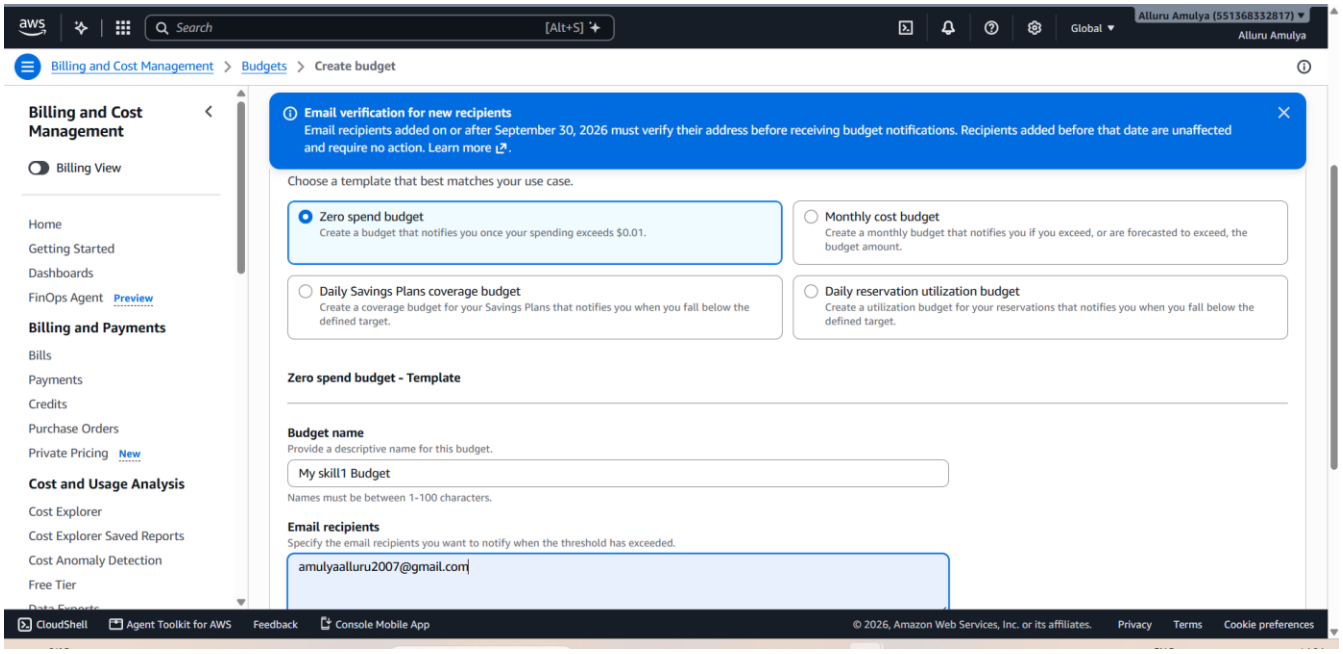
3. From the left navigation panel, select Budgets under Budgets and Planning.



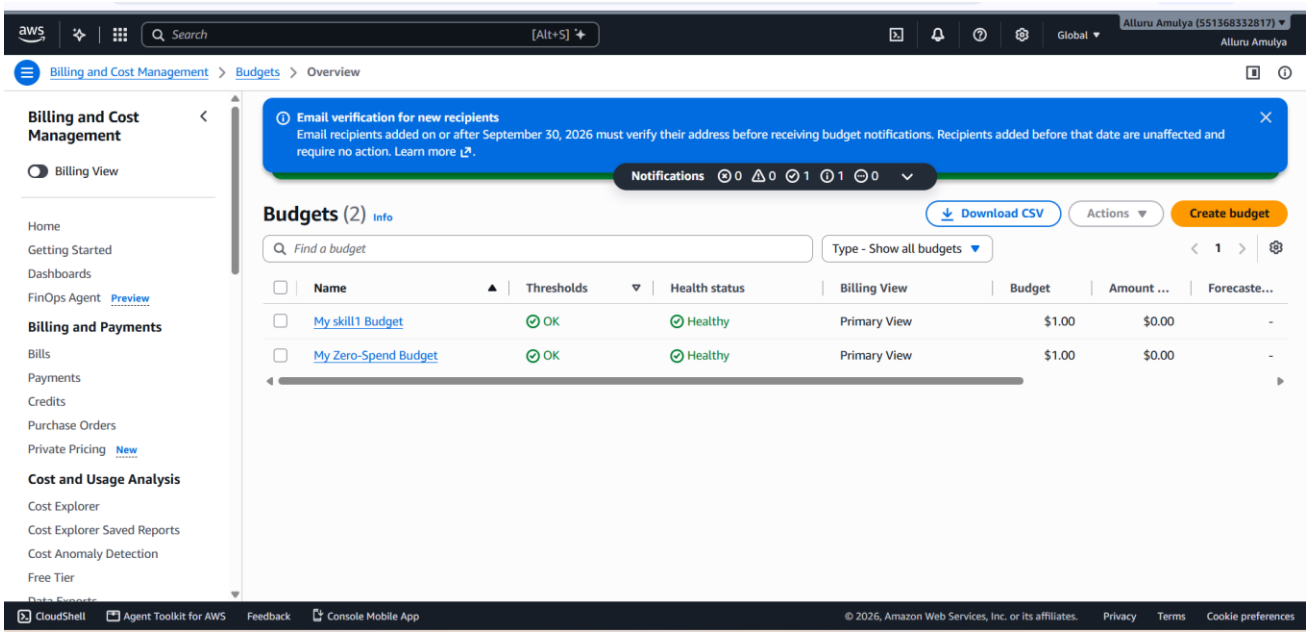
4. Click Create budget.
5. Under Budget setup, select Use a template (simplified) which is set by default, In Templates – *new select Zero spend budget (Create a budget*

that notifies you once your spending exceeds $0.01) which is set by default and then give the budget name and email id and goto step 17.

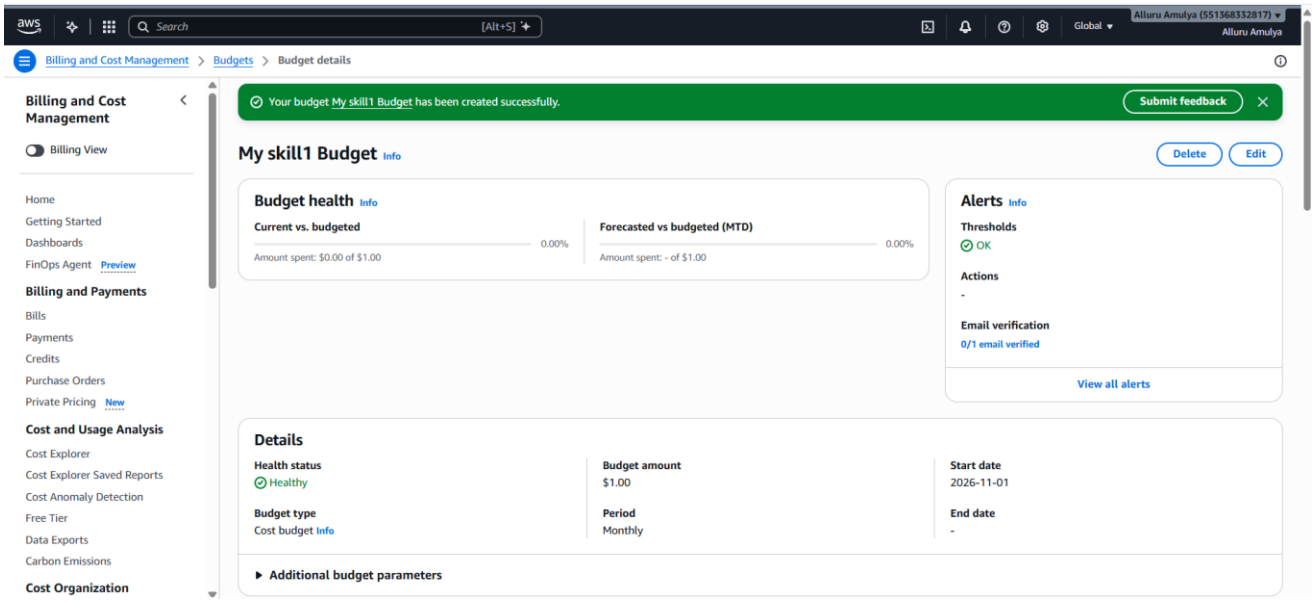




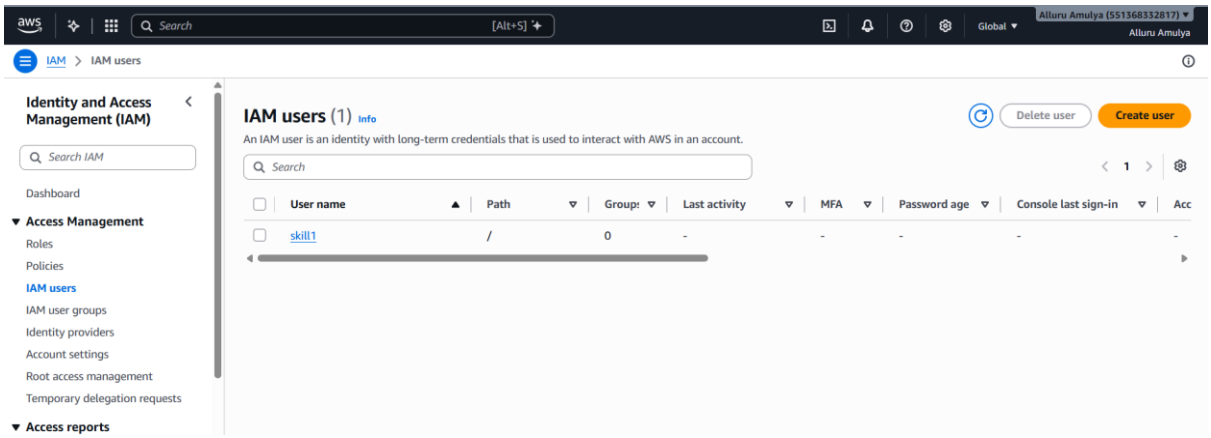
6. Verify that the newly created budget appears in the AWS Budgets dashboard.

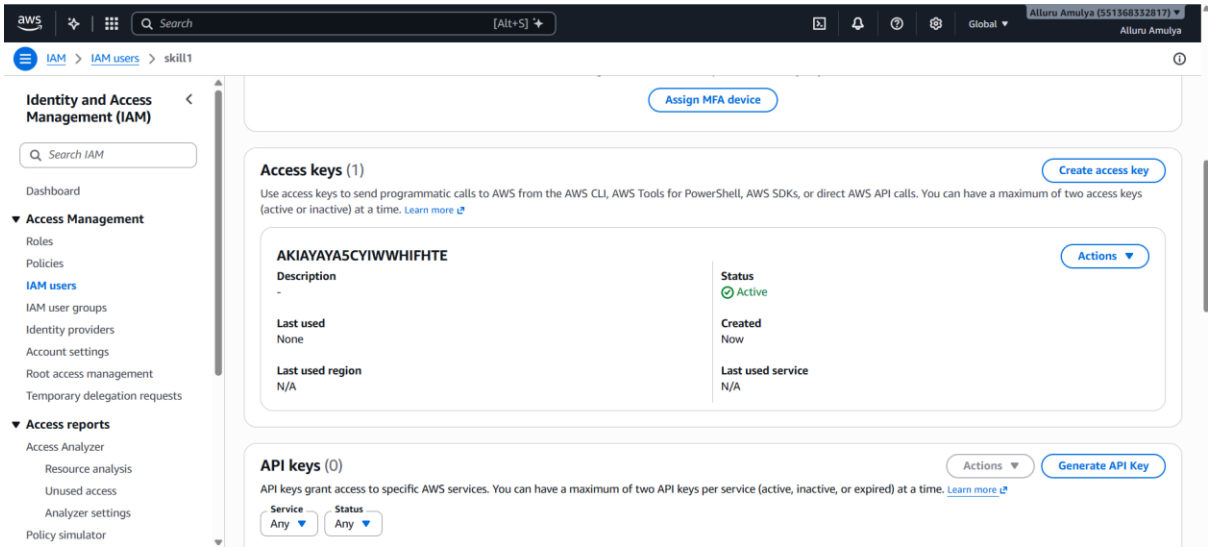


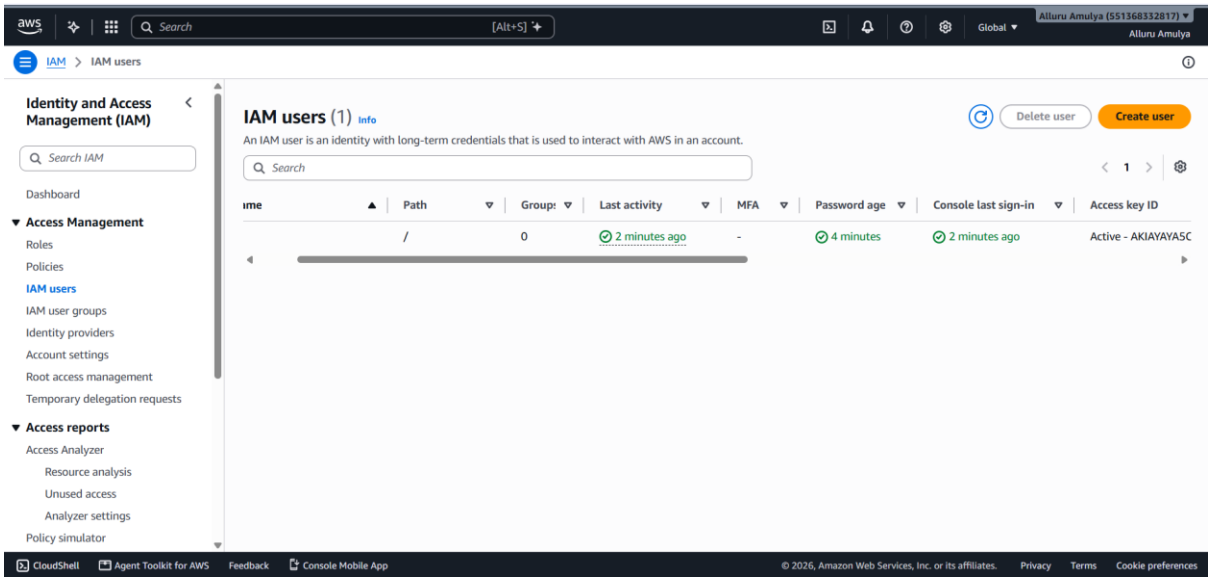
7. Check your email for AWS budget notifications when the configured thresholds are reached. Verify your email address if AWS requests confirmation.

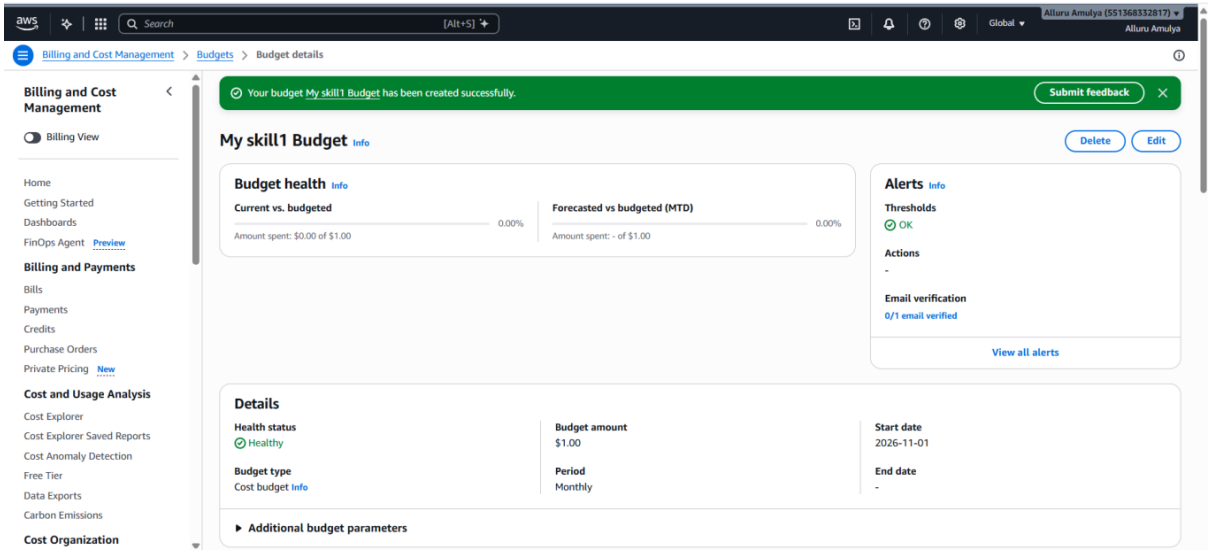


Output:


Search
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Global
Alluru Amulya (551368332817)
Alluru Amulya
IAM > IAM users
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
IAM users (1) Info
An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.
Delete user
Create user
Search
User name
Path
Group:
Last activity
MFA
Password age
Console last sign-in
skill1
/
0


Search
[Alt+S]
Global
Alluru Amulya (551368332817)
Alluru Amulya
IAM > IAM users > skill1
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IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Policy simulator
Assign MFA device
Access keys (1)
Create access key
Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. Learn more
AKIAYAYA5CYIWWHIFHTE
Actions
Description
-
Status
Active
Last used
None
Created
Now
Last used region
N/A
Last used service
N/A
API keys (0)
Actions
Generate API Key
API keys grant access to specific AWS services. You can have a maximum of two API keys per service (active, inactive, or expired) at a time. Learn more
Service
Any
Status
Any


Search
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Global
Alluru Amulya (551368332817)
Alluru Amulya
IAM > IAM users
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Create user
Search
Path
Group:
Last activity
MFA
Password age
Console last sign-in
Access key ID
/
0
2 minutes ago
4 minutes
2 minutes ago
Active - AKIAYAYA5C
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Privacy
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Search
[Alt+S]
Global
Alluru Amulya (551368332817)
Alluru Amulya
Billing and Cost Management
Budgets
Budget details
Billing and Cost Management
Billing View
Home
Getting Started
Dashboards
FinOps Agent Preview
Billing and Payments
Bills
Payments
Credits
Purchase Orders
Private Pricing New
Cost and Usage Analysis
Cost Explorer
Cost Explorer Saved Reports
Cost Anomaly Detection
Free Tier
Data Exports
Carbon Emissions
Cost Organization
Your budget My skill1 Budget has been created successfully.
Submit feedback
My skill1 Budget Info
Delete
Edit
Budget health Info
Current vs. budgeted
0.00%
Amount spent: $0.00 of $1.00
Forecasted vs budgeted (MTD)
0.00%
Amount spent: - of $1.00
Alerts Info
Thresholds
OK
Actions
-
Email verification
0/1 email verified
View all alerts
Details
Health status
Healthy
Budget amount
$1.00
Start date
2026-11-01
Budget type
Cost budget Info
Period
Monthly
End date
-
Additional budget parameters